

CHE 210: Introduction to Computing

Unique Number: 15990

Changes to the schedule and syllabus may be made at the instructor's discretion. It is your responsibility to note these changes when announced.

Instructor

Dr. Benjamin K. Keitz (he/him)
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Office Hours:
Tu 3-4 pm : CPE 4.462,
W 10-11 am : CPE 4.446

Teaching Assistant

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M 3-4 pm : EER 4.704
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Tutor

TBA
Office Hours: TBA

Lectures:

MWF : 9 am – 10 am : CPE 2.220

Please be punctual. In-person attendance and class participation is strongly encouraged. Lectures will typically consist of completing note handouts or activities (example problems). Lectures will be recorded for later upload and can be accessed through the Panopto link on Canvas. Written notes will also be uploaded to Canvas.

Recitation:

N/A

Course Goals and Overview:

This course introduces computing and calculations for chemical engineering problems as well as an overview of the basic principles for mass balances in chemical engineering. An emphasis of this course is setting up problems and defining spreadsheet-based calculations for problem solving. Overall, this course will introduce the principles and calculation techniques used in the field of chemical engineering as well as provide an exposure of the various areas and facets of current chemical engineering research. This course is designed to:

- (1) Provide you with basic knowledge of spreadsheet calculations for solving systems of equations, function minimization, and plotting.
- (2) Provide you with the fundamentals of material balances as applied to chemical engineering to enable the analysis of a chemical process.
- (3) Develop efficient methods of and life-long skills for problem solving through exercise problems and thought experiments.
- (4) Offer practice in defining problems and analyzing data.
- (5) Help you decide if you have chosen the right field.

Student Outcomes and ABET Program Outcomes Achieved:

- (1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Knowledge, Abilities, and Skills Students Should Gain From this Course:

- (1) Ability to construct basic excel spreadsheets for manipulating, plotting, and fitting data
- (2) Familiarity with excel Solver module and built-in functions (i.e. Solver and Linear Regressions)
- (3) Understanding extrapolation
- (4) Familiarity with quantities
- (5) Intuition about estimation and orders of magnitude
- (6) Exposure to flow sheets and problem solving strategies
- (7) Exposure to systems analysis and concepts of unit operations
- (8) Ability to convert between different sets of units
- (9) Ability to write and balance chemical reaction equations
- (10) Ability to define steady state and distinguish between open and closed systems
- (11) Ability to solve material balances for systems with or without a chemical reaction;
- (12) Ability to apply spreadsheet calculations to mass balance problems
- (13) Exposure to library and reference data sources
- (14) Ability to work in groups

Impact on Subsequent Courses in Curriculum:

Virtually every subsequent ChE course requires material balance concepts as well as computational skills.

Course Prerequisites:

- (1) Chemical Engineering Major
- (2) Chemistry (stoichiometry, concept of mole, molecular weight, concentration, molarity, molality, etc.; enthalpy of reaction, ideal gas law, pressure, absolute temperature) – High School Chemistry
- (3) Basic knowledge of statics and dynamics in physics (concepts of force, work, pressure, kinetic energy, potential energy) – High school physics
- (4) Time management skills to balance course load and reading assignments.

Course Textbooks:

Course material is primarily derived from two textbooks. You are not required to purchase these textbooks, but they may have example problems and additional discussion that you may find helpful. Electronic versions are also available. Finally, it is worth noting that these textbooks will be useful (and might be required) in subsequent courses.

Felder & Rousseau (3rd or 4th edition) Elementary Principles of Chemical Processes

Ronald W. Larsen (5th Edition), Engineering with Excel. Person, 2017.

Additional Resource Requirements:

You are required to have the following:

Laptop Computer with Microsoft Office as per latest UT Engineering standards

<http://che.utexas.edu/current-students/undergraduate-students/undergraduate-office/laptop-requirements/>

The UT Instapoll app will be used for in class participation questions. You will need an electronic device (phone, laptop, iPad) with access to Canvas to complete these questions.

Course Website: A course page will be maintained on Canvas — Please check it regularly!

Grading:

Course Grades will be based on the following grading scheme:

<u>Course Grading</u>	
Exams (best 3 of 4)	65%
Quizzes	15%
Project	15%
Participation	5%

The following grade cutoffs will be used: A = >93; A- = 90-93; B+ = 87-90; B = 83-87; B- = 80-83; C+ = 77-80; C = 73-77; C- = 70-73; D = <70; F = <60. Grades will be curved as necessary and plus/minus grading will be used in this course.

Participation: In person attendance is strongly encouraged. There will be unannounced short problems administered through UT Instapoll at least once a week. These problems will not be graded but completing them will count toward your participation grade.

Homework and Quizzes: Homework will be assigned on Wednesdays via Canvas. A quiz containing a problem similar to one or two questions from the homework will be given the following Wednesday (a week after the assignment). Quizzes will be administered remotely through Gradescope and will require scanning and uploading written solutions. Quizzes will have a time limit (usually 45-60 min) and can be taken anytime within an ~12 hour window. For example, you can start the quiz at 10 pm and turn it in by 10:45 pm or start it at 7:30 pm and turn it in by 8:15 pm. You are strongly encouraged to work in small groups to discuss and solve the homework problems prior to the quiz, but quizzes must be completed individually. If you do not complete the homework, you will not be prepared for the quiz.

Exams: Three in-class exams are scheduled on **February 12, 2025, March 12, 2025** and on **April 16-18, 2025**. **There will be no make-up exams offered! Exams will take place in-person during the normal lecture period.** Please double check for potential conflicts with exams in other classes and let the instructor know as soon as possible. Optional review sessions (outside of class time) will be held before each of the exams and will be announced through Canvas. The final exam for this course is optional and has been set by the registrars office - **May 1, 1–3 pm**. The location of the final exam will be announced the last week of class. If you choose to take the Final Exam, the best 3/4 Exam scores will count toward your final grade.

Project: The project is due the final class day and should consist of a process flow diagram and associated description for a complex, real-world process. You may work in groups of 3-4 students. The process for your project must be approved by the instructor in writing (email is fine). If you decide to change your process, the new topic must also be approved. Potential processes to describe are: “N95 Mask Production”, “COVID-19 Vaccine Production”, “Local NH₃ (ammonia) Production”, “Photovoltaic Production”, “Polyethylene Production”. Your final project should con-

sist of the following:

- (1) A detailed process flow diagram describing your chosen process. This should be prepared in Powerpoint or another computer program. All streams should be numbered and stream compositions provided as best as possible. Unit operations should also be clearly labelled. You will likely need to research your process to complete this diagram.
- (2) A written description of your process. This should be 3-4 pages in length and should include assumptions, limitations, and potential improvements to your process flow diagram.
- (3) Mass balance calculations for one of the units in your process with at least 3 streams or components (in other words, a non-trivial balance). You may make educated assumptions as needed to come up with reasonable numbers, as long as they are justified in the written description.
- (4) A description of the contribution of each member in your group (if working as a team). This should be about a paragraph in length. Note that team members may receive different grades if the workload was clearly unequal.
- (5) A description of changes made because of peer-review and discussions with the TA/instructor.

There will be several smaller, project-related assignments throughout the semester to help you make progress and give you feedback. These are summarized below:

Assignment	Description	Due Date
Topic Selection	One paragraph written description justifying topic selection and listing team members	02/07/25
PFD Draft	Rough draft of process flow diagram	03/07/25
Rough Draft	Rough Draft including written description, and references	04/05/25
Peer-Review	Provide feedback on projects from other teams	04/11/25
Final Project		04/28/25

Regrade Requests: You have one week after an assignment is returned to you to request a re-grade. Re-grading requests should be submitted in writing at the end of the lecture period and should consist of a few sentences on a *separate* page that describe your rationale for why a re-grade is necessary. Nothing is to be marked on the original assignment or else the re-grade request will be automatically voided. Regrade requests may also be submitted via Gradescope.

Late Work: Assignments must be turned in by the assigned due date. Late work will have 20% deducted for every 24 hours past the original due date. For example, if an assignment is due Wednesday at midnight, submitting it Thursday at 1 am would result in 20% off, submitting it Friday at 1 am would result in 40% off, etc. Assignments turned in more than 48 hours past the original due date will not be accepted.

Absence: No excuses for missed exams, or quizzes will be accepted other than written certified medical excuses or written letters on university letterhead for UT-related school activities. By UT Austin policy, you must notify the instructors of your pending absence at least fourteen days prior to the date of observance of a religious holy day. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given an opportunity to complete the missed work within a reasonable time after the absence.

Academic Integrity:

The University has a strict policy on academic integrity. Any form of plagiarism or academic dishonesty will NOT be tolerated in ChE 210. If you have any questions, please consult the websites below:

http://deanofstudents.utexas.edu/sjs/acint_student.php
http://www.utexas.edu/courses/clubmed/plgrism_cit.html

In particular, students are obliged to follow the Student Honor Code that states “As a student of The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity.”

Failure to comply with the University’s policy will result in a “zero” on the assignment and an “F” for CHE 210.

Q Drop Policy:

If you want to drop a class after the 12th class day, you’ll need to execute a Q drop before the Q-drop deadline, which typically occurs near the middle of the semester. Under Texas law, you are only allowed six Q drops while you are in college at any public Texas institution. For more information, see:

<http://www.utexas.edu/ugs/csacc/academic/adddrop/qdrop>

Sharing of Course Materials is Prohibited:

No materials used in this class, including, but not limited to, lecture hand-outs, videos, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have the instructors explicit, written permission. Sharing course materials for the purpose of incorporating them into machine learning models is also prohibited. Unauthorized sharing of materials promotes cheating. It is a violation of the University’s Student Honor Code and an act of academic dishonesty. I am aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in sanctions, including failure in the course.

Class recordings are reserved only for students in this class for educational purposes and are protected under FERPA. The recordings should not be shared outside the class in any form. Violation of this restriction by a student could lead to Student Misconduct proceedings.

Use of Generative AI

Students are invited to use AI platforms, such as ChatGPT, Gemini, etc. to help prepare for assignments and projects (e.g., to help with brainstorming). You are also welcome to use AI tools to help revise and edit your work (e.g., to help identify flaws in reasoning, spot confusing or underdeveloped paragraphs, or to simply fix citations). When submitting work, students must clearly identify any writing, text, or media generated by AI. This can be done in a variety of ways. In this course, parts of essays generated by AI should appear in a different colored font, and the relationship between those sections and student contributions should be discussed in the final project submission. You are also welcome to use generative AI platforms for developing Python, MATLAB, or Julia code. This can be particularly helpful if you are struggling with syntax or encountering bugs in your code. If you are relying on these services, please note that they are web-based and may not be available during an exam.

Emergency Preparedness and Emergency Plan Instructions:

Please take note of all emergency exits, exit procedures, and University Emergency Communications Policies for the lecture and recitation buildings. Students requiring assistance in evacuation should inform the Professor of the course in writing of their needs during the first week of class so that this information is properly transmitted to the Fire Prevention Services office. In the case of a COVID-19 outbreak, the course will transition to a completely remote format for at least 10 days.

Americans with Disabilities Act:

The University of Texas at Austin is on record as being committed to both the spirit and letter of federal equal opportunity legislation; reference Public Law 92-112 - The Rehabilitation Act of 1973 as amended. With the passage of federal legislation entitled *Americans with Disabilities Act (ADA)*, pursuant to section 504 of the Rehabilitation Act, there is renewed focus on providing this population with the same opportunities enjoyed by all citizens.

As a faculty member, I am required by law to provide “reasonable accommodations” to students with disabilities, so as not to discriminate on the basis of that disability. Student responsibility primarily rests with informing the Professor of their need for accommodation and in providing authorized documentation through designated administrative channels. Information regarding specific diagnostic criteria and policies for obtaining academic accommodations can be found at <https://sites.utexas.edu/disability/disability-resources/>. Students with accommodations will be able to register for alternate exam times/lengths through the [Student Testing Center](#).

Title IX Reporting:

Title IX is a federal law that protects against sex and gender-based discrimination, sexual harassment, sexual assault, sexual misconduct, dating/domestic violence and stalking at federally funded educational institutions. UT Austin is committed to fostering a learning and working environment free from discrimination in all its forms. When sexual misconduct occurs in our community, the university can: 1. Intervene to prevent harmful behavior from continuing or escalating. 2. Provide support and remedies to students and employees who have experienced harm or have become involved in a Title IX investigation. 3. Investigate and discipline violations of the university’s relevant policies (<https://titleix.utexas.edu/relevant-policies/>). Beginning January 1, 2020, Texas Senate Bill 212 requires all employees of Texas universities, including faculty, report any information to the Title IX Office regarding sexual harassment, sexual assault, dating violence and stalking that is disclosed to them. Texas law requires that all employees who witness or receive any information of this type (including, but not limited to, writing assignments, class discussions, or one-on-one conversations) must be reported. I am a Responsible Employee and must report any Title IX related incidents that are disclosed in writing, discussion, or one-on-one. Before talking with me, or with any faculty or staff member about a Title IX related incident, be sure to ask whether they are a responsible employee. If you would like to speak with someone who can provide support or remedies without making an official report to the university, please email advocate@austin.utexas.edu. For more information about reporting options and resources, visit <http://www.titleix.utexas.edu/>, contact the Title IX Office via email at titleix@austin.utexas.edu, or call 512-471-0419.

Although graduate teaching and research assistants are not subject to Texas Senate Bill 212, they are still mandatory reporters under Federal Title IX laws and are required to report a wide range of behaviors we refer to as sexual misconduct, including the types of sexual misconduct covered under Texas Senate Bill 212. The Title IX office has developed supportive ways to respond to a survivor and compiled campus resources to support survivors.

Personal Pronouns:

Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with differences of race, culture, religion, politics, sexual orientation, gender, gender variance, and nationalities. Class rosters are provided to the instructor with the student’s legal name, unless they have added a “preferred name” with the Gender and Sexuality Center (<http://diversity.utexas.edu/genderandsexuality/publications-and-resources/>). I will gladly honor your request to address you by a name that is different from what appears on the official roster, and by the gender pronouns you use (she/he/they/ze, etc). Please advise me of any changes early in the semester so that I may make appropriate updates to my records.

Approximate Course Outline (subject to change)

Week	Day	Date	Lecture/Excel Clinic	Assignments
1	M	1/13/25	First Class Day: Course Intro Presentation and Syllabus Overview	

	W	1/15/24	Lecture 1: Review of Units and Dimensions	Homework 1 assigned
	F	1/17/25	Excel Clinic 1: Excel Basics, Graphic, Error Bars	
2	M	1/20/25	No Class - MLK Jr Day	
	W	1/22/25	Lecture 2: Guesstimation Continued + Interpolation/Extrapolation	Quiz 1, Homework 2 Assigned
	F	1/24/25	Excel Clinic 2: Unit Conversions in Excel	
3	M	1/27/25	Lecture 3: Guesstimation and Unit Conversion Example Problems	
	W	1/29/25	Lecture 4: Everything is a Process	Quiz 2, Homework 3 Assigned
	F	1/31/25	Excel Clinic 3: More Unit Conversions and Linearization	
4	M	2/3/25	Lecture 5: Reading Process Flow Diagrams	
	W	2/5/25	Lecture 6A: Drawing Process Flow Diagrams	Quiz 3, Homework 4 Assigned
	F	2/7/25	Excel Clinic 4: Linearization and Interpolation/Extrapolation	Project Topic Selection & Groups due
5	M	2/10/25	Lecture 6B: Drawing Process Flow Diagrams	
	W	2/12/25	Exam 1	Homework 5 Assigned
	F	2/14/25	Lecture 7A: Process Flow Variables	
6	M	2/17/25	Lecture 7B: Process Flow Variables (wet v dry basis)	
	W	2/19/25	Lecture 8: Process Flow Variable Problems	Quiz 5, Homework 6 Assigned
	F	2/21/25	Excel Clinic 5: Linear Algebra and Matrix Operations	
7	M	2/24/25	Lecture 9A: Intro to Material Balances	
	W	2/26/25	Lecture 9B Intro to Material Balances	Quiz 6, Homework 7 Assigned
	F	2/28/25	Lecture 10A: Material Balance Example Problems	
8	M	3/3/25	Lecture 10B: Material Balance Example Problems	
	W	3/5/25	Lecture 11: Introduction to Atomic Balances	Quiz 7, Homework 8 Assigned

	F	3/7/25	Excel Clinic 6: Introduction to Solver	PFD Rough Draft Due
9	M	3/10/25	Lecture 12: Atomic Balance Examples	
	W	3/12/25	Exam 2	Homework 9 Assigned
	F	3/14/25	Excel Clinic 7: More Solver Practice	
10	M	3/17/25	Spring Break (No Class)	
	W	3/19/25	Spring Break (No Class)	
	F	3/21/25	Spring Break (No Class)	
11	M	3/24/25	Lecture 13A: Comparing Solver and Matrix Solutions	
	W	3/26/25	Lecture 13B: Comparing Solver and Matrix Solutions	Quiz 9, Homework 10 Assigned
	F	3/28/25	Excel Clinic 8: Comparing Solver & Matrices	
12	M	3/31/25	Lecture 14A: Single Unit Systems and Solver	
	W	4/2/25	Lecture 14B: Single Unit Systems and Solver	Quiz 10, Homework 11 Assigned
	F	4/4/25	Excel Clinic 9: More Solver Practice	Project Rough Draft due
13	M	4/7/25	Lecture 14: Single Unit Systems and Solver	
	W	4/9/25	Lecture 15A: Intro to Multi-Unit Systems	Quiz 11, Homework 12 Assigned
	F	4/11/25	Excel Clinic 10: Solver and Multi Unit Systems	Peer Review of Project Rough Drafts Due
14	M	4/14/25	Exam 3 Review	
	W	4/16/25	Exam 3 Part 1	
	F	4/18/25	Exam 3 Part 2	
15	M	4/21/25	Lecture 15B: More Multi-Unit Systems	
	W	4/23/25	Lecture 15C: More Multi-Unit Systems	
	F	4/25/24	Self-Study: Work on Final Projects	
16	M	4/28/25	Last Class Day : Course wrap up	Final Project Due Date